

ON
THE INACCURACIES
WHICH PROBABLY EXIST IN THE
CENSUS RETURNS OF AGES.

BY
THOMAS A. WELTON, F.S.S.

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T. BRAKELL, PRINTER, COOK STREET.
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ON THE INACCURACIES WHICH PROBABLY EXIST IN THE CENSUS RETURNS OF AGES.

By Thomas A. Welton, F.S.S., &c.

SOME years ago, before the publication of the third volume of the Census of England and Wales, 1871, I was anxious to calculate the approximate death rates in particular parts of England, amongst females at certain ages, during each of the years 1851–1870. I therefore applied myself to the preliminary process of estimating the population in each year at the ages in question.

I found that it was not safe to assume that, if population in a given county or district had increased generally, at a certain rate, there had been an increase at each particular age at something like the same rate. On the contrary, there was a surprising irregularity in the respective rates of increase or decrease observed amongst persons of the several ages in the same population.

Table I shows the rates of increase in the numbers enumerated at each age up to 80 amongst males and females respectively in the several registration divisions, and in England and Wales. On inspecting this table, it will be seen that, partly owing to the effect of emigration, the ratios of increase at particular ages are very diversified. In the Eastern Counties males aged 20–25 decreased by nearly one-tenth, but males aged 65–70 increased by nearly a quarter. In the same division, females aged 25–30 decreased by almost 5 per cent., whilst females aged 40–45 increased by more than 15 per cent.

Such being the state of the facts, I abandoned all thought of deducing from the total numbers enumerated in 1871 any reliable estimates of the numbers at particular ages, unless indeed some other mode of treatment of the figures could be found leading to more regular results.

On comparing with the population enumerated in 1851 the numbers, ten years older, found to be living in 1861, I obtained other sets of ratios, which are shown in Table II.

The following is a comparison between the proportions for England and Wales, shewn in Table II, and the numbers out of 100 living in 1851 who would have survived in 1861 according to the English Life Table No. 3.

Ages in 1851.	Proportion of survivors out of 100 living in 1851.			
	By the Census.		By the Life Table.	
	Males.	Females.	Males.	Females.
0–5 ⁽⁴⁾	90.1	89.2	86.4	86.7
5–10	91.2	93.5	94.2	94.0
10–15	89.2	102.1	93.7	93.4
15–20	84.1	94.4	92.0	91.6
20–25	83.2	83.2	90.9	90.4
25–30	84.4	82.3	89.9	89.6
30–35	89.2	88.6	88.6	88.6
35–40	85.1	85.9	86.8	87.4
40–45	82.3	83.8	84.3	85.9
45–50	76.1	77.6	80.5	83.2

50–55	76.7	80.2	75.3	78.2
55–60	68.9	74.1	67.7	70.7
60–65	56.5	60.2	56.9	60.3
65–70	47.3	50.5	43.5	47.2

The emigration of females of English birth has by no means been on an insignificant scale, at any time since 1851; but owing to the immigration of large numbers of women from Ireland, Scotland, and foreign parts, the net loss has been only moderate. We should therefore be justified in looking for a certain correspondence between the proportions of females who might be expected to survive, according to the Life Table, and those proportional figures which represent a comparison of the census figures of 1861 with those of 1851. In point of fact, we discover that, at ages 10–20, the proportions who should survive the next ten years, according to the Life Table, are far exceeded by those who apparently do survive; and this state of things in the next ten years is reversed. So that the figures suggest a faulty return of the ages of the female population, exaggerating the numbers aged 20–30, and perhaps depressing those aged 30–40.

Again, we have reason to believe that some of the children under five years old are returned as being fully of that age, whilst next to none who have attained five years of age would be returned as being younger. This would account for the dissimilarity of the ratios of survivors at the earliest period of life.

By assuming, as experimental suppositions, that the proportionate errors in the censuses of 1851 and 1861 at each age were equal, and that female emigration and immigration neutralised each other, I obtained a set of corrections of the census enumerations of females which indicated that there was a tendency in the case of young persons under twenty to exaggerate their ages, but that women aged 25–30, and at each successive quinquennial period of life, at least up to 55–60, had a tendency to understate their ages. There was likewise a certain disposition to return ages in round numbers of years, though this was far less remarkable than in the Irish census of 1851, from which the following figures are taken.

Age returned.	Males.	Females.
35 and under 40	150,471	165,966
40 „ 45	187,410	217,986
45 „ 50	109,618	117,345
50 „ 55	156,337	176,782
55 „ 60	73,511	79,111
60 „ 65	100,963	130,740

The understatement, at the ages mentioned, appeared to have been no greater in this country than would have resulted from *every woman*, aged 25–60, calling herself one year younger than her true age.

It will be obvious how great the utility of censuses taken at intervals of *five* years would be, in enabling us to measure more accurately the results of these tendencies to mis-state ages, and particularly the effect of ages being returned in round numbers.

The corrections applicable to the returns of males could not be even approximately determined without taking into account the effects of emigration upon the numbers of that sex. And as the rates of mortality are subject to considerable variation, year by year, I came to the conclusion that more reliable results must be sought by the aid of—

- (1) Estimates of the numbers of the *recorded* deaths which happened amongst persons *born* in each quinquennial period. ^[6]
- (2) A computed allowance for unregistered births in excess of the unregistered deaths of infants.
- (3) Estimates of the loss or gain of population through migrations at each age.

Table III exhibits the results obtained by computing the first and second of these elements, and showing the remaining difference between the two sets of census figures, as the combined result of migrations and errors.

On examining this table, I thought it extremely probable that the numbers of deaths at high ages were exaggerated, because I know of no reason for supposing that the census numbers at such ages are less than the truth; and if they are equal to or more than the real numbers of the living, we are compelled to conclude either that there is a considerable immigration of old people, or, what seems much more compatible with such knowledge as we possess, a tendency exists to exaggerate the ages both of the living and the dying amongst those who are over seventy years old.

By the aid of the "English Life Table No. 3" it was ascertained that in a population resulting from births increasing at 1 per cent. per annum, the following numbers would represent the proportions of persons living and dying at high ages:—

Ages.	Living.		Ages.	Living.		Proportion of the former to the latter.	
	Males.	Fem'l's.		Males.	Fem'l's.	Males. as 100 to	Fem'l's. as 100 to
75 & under 80	373054	428741	74½ & under 79	369162	420783	99.0	98.1
80 „ „ 85	174287	213540	79 „ „ 84	207496	250662	119.1	117.4
85 „ „ 90	59641	79253	84 „ „ 89	76091	99340	127.6	125.3
90 „ „ 95	13652	20037	89 „ „ 94	19023	27331	139.3	136.4
95 „ „ 100	1887	3119	94 „ „ 99	2934	4728	155.5	151.6
100 & upwards.	145	279	99 and upwards	260	484	179.3	173.5
Ages.	Deaths.		Ages.	Deaths.		Proportion of the former to the latter.	
	Males.	Fem'l's.		Males.	Fem'l's.	Males. as 100 to	Fem'l's. as 100 to
75 & under 85	78695	84957	74½ & under 84	79184	84673	100.6	99.7
85 „ „ 95	19617	24868	84 „ „ 94	24024	29893	122.5	120.2
95 & upwards	932	1476	94 and upwards	1393	2152	149.5	145.8

From the above table it may be deduced that, supposing persons aged 71–75 call themselves (or are described as being) on an average six months older than their true age, and if after 75 the exaggeration averages an entire year, a very great impression must be thereby made upon the returns.

Having arrived thus far, I thought it would be convenient in the first instance to try whether these suggestions, which I imagine will be deemed moderate and probable, would suffice to explain the apparent influx of aged persons, shown in Table III; and whether other suppositions, not less reasonable, would serve to overcome the remaining difficulties which appear on the face of that table.

The female population being least disturbed by migrations, the necessary calculations were made in relation to it, before proceeding further. First, the deaths shewn in Table III had to be modified, as follows:—

Born in the years	Deaths of Females as in Table III.		As now Corrected.	
	1851–60.	1861–70.	1851–60	1861–70.
1801–05	65030	97481	65030	98802
1796–1800	72028	108636	72028	112636
1791–95	82975	114233	84226	118269
1786–90	93843	105704	97626	107225

1781–85	99612	78080	103438	76172
1776–80	94274	43589	95728	39172
1771–75	71487	17466	69678	14330
1766–70	40514	4849	36401	3732
1761–65	16604	894	13613	617
1756–60	4724	73	3631	50
1751–55	921	—	636	—
1746–50	76	—	53	—
Totals	642088	571005	642088	571005

Then it was necessary to make some assumptions as to the effect of migrations of females into and from this country, and the following estimates were provisionally adopted:—^(8a)

ESTIMATED AVERAGE ANNUAL NET RESULT OF MIGRATIONS.					CONSEQUENT ESTIMATE OF NET LOSS OR GAIN IN 10 YEARS. ^(8a)				
Age (at end of the year of migration)	1851–60.		1861–70.		Age (at the end of the decennium)	1851–60.		1861–70.	
	Immigrants. ^(8a)	Emigrants. ^(8b)	Immigrants. ^(8a)	Emigrants. ^(8b)		Net loss.	Net gain.	Net loss.	Net gain.
0–5	2000	3250	1550	2675	0–5	3750	—	3375	—
5–10	2000	3000	1530	2450	5–10	9250	—	8385	—
10–15	2800	2280	2170	1900	10–15	6000	—	6040	—
15–20	5300	3150	4200	2740	15–20	—	7000	—	3890
20–25	2400	3900	1860	3200	20–25	—	7300	—	3820
25–30	1050	2700	800	2190	25–30	8150	—	7950	—
30–35	630	1630	500	1340	30–35	14250	—	12150	—
35–40	400	1050	320	870	35–40	10250	—	8630	—
40–45	390	690	320	590	40–45	6150	—	5240	—
45–50	280	520	210	420	45–50	3500	—	3080	—
50–55	190	420	145	345	50–55	2500	—	2190	—
55–60	100	250	75	200	55–60	2000	—	1795	—
Totals	17540	22840	13680	18920	60–65	1200	—	1025	—
					65–70	300	—	250	—

No great confidence can be placed in these last calculations as to the effect of migrations at particular ages. The facts bearing on the subject preserved in official records with which I am acquainted are but scanty. The rough, general idea which may be gathered from the table does, however, approximate more or less closely to the truth, and may be usefully contrasted with the violent fluctuations indicated in Table III. These shew, in the midst of their extravagance, a kind of regularity at particular ages, thus—

Age at end of Decennium.	1851–60.		1861–70.	
	Loss.	Gain.	Loss.	Gain.

0-5	30575	...	42643	...
5-10	...	3937	...	15075
10-15	...	24995	...	29722
15-20	7416	...	14314	...
20-25	...	85027	...	73654
25-30	...	27678	...	42046
30-35	69827	...	74590	...
35-40	63559	...	54880	...
40-45	4438	...	7513	...
45-50	11175	...	11472	...
50-55	16118	...	18811	...
55-60	26073	...	28220	...
60-65	...	35	1360	...
65-70	...	12614	...	18345
70-75	7310	...	7982	...

Finding it quite impossible to believe in the successive migrations which must have taken place, if this extract from Table III represented the truth, I pursued the enquiry as to what is the alternative of such a belief.

Proceeding to calculate what corrections must be made in the census returns of the numbers of females, if the amended estimates of losses by death and the calculated results of migrations be adopted, I first assumed that each of the three censuses of 1851, 1861, and 1871 might be rendered erroneous by misstatements as to ages in *fixed proportions* at the several periods of life.

The endeavour to find such a fixed scale of proportions as would rectify all the censuses was, however, unsuccessful; and it became evident that the deviations from the truth were greatest in 1851 and least in 1871. Three scales of proportions were then arrived at empirically, embodying the idea of diminishing degrees of error. In the course of the researches made for the sake of adjusting these scales, I found reason to believe that the allowances for unregistered births in the years 1856-60 and 1861-65 should be increased by $\frac{1}{2}$ per cent. That addition having been made to the estimated numbers of births, it became requisite that equal numbers should be added to the estimated losses by emigration; and by finally amending the three scales of proportions in conformity with these alterations, the results shewn in Table IV were at last arrived at.

These results require us to believe that, whilst our calculation of the effect of migrations in 1851-60 was near the truth, the similar estimate for 1861-70 was rather beside the mark. The numbers now required to be substituted, however, appear to me to be acceptable, especially when we have regard to our inability to form any opinion as to the ages of those persons of English birth who returned in large numbers from the United States about the time of the war of secession.

The next thing to do was to ascertain what transpositions of the census figures are involved, in case we accept the numbers shewn in Table IV as being approximately correct.

Table V exhibits these transpositions, and it will be seen that they lead to the conclusions shewn in the following statement:—

STATEMENT A.—Shewing the proportions (per cent.) of female population returned at each age, who were really older or younger than represented.

Returned as	1851. انقلاب	1861.	1871.
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aged	Really older.	Age correct.	Really y'nger.	Really older.	Age correct.	Really y'nger.	Really older.	Age correct.	Really y'nger.
0-5	...	100.0	...	...	100.0	...	...	100.0	...
5-10	...	97.0	3.0	...	97.1	2.9	...	97.3	2.7
10-15	...	98.1	1.9	...	98.1	1.9	...	98.2	1.8
15-20	...	97.4	2.6	...	97.9	2.1	...	98.0	2.0
20-25	4.7	92.3	3.0	5.3	92.4	2.3	4.8	93.2	2.0
25-30	9.8	90.2	...	10.3	89.7	...	9.5	90.5	...
30-35	13.3	86.7	...	13.5	86.5	...	11.5	88.5	...
35-40	12.6	87.4	...	12.6	87.4	...	11.9	88.1	...
40-45	16.5	83.5	...	16.1	83.9	...	15.3	84.7	...
45-50	15.1	84.9	...	14.6	85.4	...	13.9	86.1	...
50-55	16.5	83.5	...	16.5	83.5	...	15.5	84.5	...
55-60	8.4	91.6	...	8.2	91.8	...	7.2	92.8	...
60-65	9.0	91.0	...	8.8	91.2	...	8.1	91.9	...
65-70	1.7	98.3	...	2.0	98.0	...	1.8	98.2	...
70-75	...	100.0	...	.3	99.7	...	.5	99.5	...
75-80	...	99.9	.1	...	100.0	...	...	100.0	...
80-85	...	91.6	8.4	...	92.6	7.4	...	93.2	6.8
85-90	...	90.5	9.5	...	92.1	7.9	...	93.1	6.9
90-95	...	84.2	15.8	...	85.6	14.4	...	88.7	11.3
95-100	...	62.2	37.8	...	62.5	37.5	...	68.1	31.9
100 & up.	...	38.0	62.0	...	38.4	61.6	...	50.4	49.6

The ratios in the above statement signify that at no time of life does the apparent *under statement* of age average so much as an entire year; ^(طابقاً) and the exaggeration even at high ages also appears to average less than a year, so that there is, I think, no such unlikelihood about the figures as should lessen their credibility. When the irregular results of Table III for females are compared with the much more probable results shewn in Table IV, and the assumptions by which the latter were arrived at, and through the adoption of which the registered births and deaths, the returns at successive censuses of the numbers of the female population at several ages, and the computed losses by migration have been brought into close agreement, are considered, I think it will be seen that it is very much safer to adopt my corrections than to rely on the actual returns.

As the operation by which the age-returns are to be corrected is essentially one of transposition, I suppress the ratios upon the footing of which I constructed Table IV, and would employ the proportions shewn in Statement A, in applying similar corrections to the population returns in detail. In the absence of any means of judging what variations there may be in different parts of the country in the extent of the misrepresentations as to ages, I should be inclined to make use of these proportions in every case, though not without apprehending that inaccuracies of some consequence may thus be fallen into.

One more test may be applied before we finally adopt the figures exhibited in Table IV as representing (very nearly) the true female population. The ratios borne by the population there shewn to exist in 1861 and 1871 respectively, *plus* emigrants, to the numbers ten years earlier, may be computed and compared with those already shewn, which were derived from the English Life Table No. 3, thus:—

Age at the end of the ten years.	Proportion of survivors (including emigrants) out of 100 Females who were living 10 years earlier.		The like proportion, according to the English Life Table, No. 3.
	According to corrected figures, 1851 and 1861.	According to corrected figures, 1861 and 1871.	
10–15	87.4	87.5	86.7
15–20	94.2	94.7	94.0
20–25	93.2	93.7	93.4
25–30	91.4	91.9	91.6
30–35	90.5	91.0	90.4
35–40	90.0	90.1	89.6
40–45	89.0	89.1	88.6
45–50	88.3	88.2	87.4
50–55	86.8	86.8	85.9
55–60	84.7	84.6	83.2
60–65	80.2	79.5	78.2
65–70	72.7	72.4	70.7
70–75	61.6	61.3	60.3
75–80	47.1	46.9	47.2

On examining the ratios thus obtained, it will be seen that they harmonise well with the probability shewn by the Life Table. Each set of ratios is symmetrically graduated, whilst the proportions obtained by the use of the uncorrected Census Tables were, as has already been observed, by turns exaggerated and depressed.

Having thus arrived at a sufficiently near approximation to the truth in the case of females, it remains for us to endeavour to do the same in that of males. The effect of emigration must be first computed, which may be done thus:—

Age at end of the year of migration.	1851–60.		1861–70.	
	Immigrants from Ireland, &c.	English Emigrants.	Immigrants from Ireland, &c.	English Emigrants.
0–5	2000	3500	1500	2200
5–10	2000	3500	1500	2200
10–15	3000	3000	2200	2000
15–20	6000	6500	4200	4000
20–25	2500	9000	1750	6000
25–30	1000	8000	750	5000
30–35	500	5000	400	3400
35–40	500	3000	350	2100
40–45	400	1850	260	1200
45–50	300	1300	200	800
50–55	200	1000	150	600

55-60	100	700	100	440
Totals	18500	46350	13360	29940

The above figures are based on an augmented estimate of unregistered male births, to correspond with the increased estimate of unregistered female births used in compiling Table IV.

The total loss or gain of male inhabitants at each age, resulting from the above migrations, might possibly have been as under, assuming the course of events to have been quite unvarying year by year:—

Age at close of decennium.	Net Loss.	
	1851-60.	1861-70.
0-5	4500	2100
5-10	12000	5600
10-15	10500	4300
15-20	4500	200 ⁽¹⁴⁾
20-25	22000	11350
25-30	54500	33600
30-35	61500	38750
35-40	44000	28750
40-45	25850	17570
45-50	15250	10000
50-55	10300	6230
55-60	7800	4470
60-65	4600	2600
65-70	1200	680
Totals	278500	165800

As, however, those who departed in the earlier years of the decennium 1861-70 for the United States, may, to a great extent, have returned home, as well as many of those who had previously left the country, it seems natural to suppose that the emigrants of 1861-70 were, on the whole, much younger than the above Table would shew.

The result of a careful study of the various figures has been to convince me that the ages of male children, as well as those of female children, are overstated. At the age 15 and under 20 males do not appear to exaggerate their ages so often as females do. From 25 to 70 there seems to be a general but slight tendency to understate age, to the extent, on an average, of a quarter or at most nearly half a year. After 70, the same tendency to exaggerate age, which was noticed in the case of females, seems to prevail.

In Tables VI and VII are contained the same approximate facts respecting males, as are furnished by Tables IV and V for our female population. The figures of all these statements have been arrived at empirically, but so as to involve the most moderate and regular corrections which will meet the difficulties of the case.

The resulting proportions of supposed accurate and inaccurate returns at each age may be thus shewn:—

STATEMENT B.

Shewing the proportions (per cent.) of male population returned at each age, who were really older or younger than represented.

Returned as aged	1851.			1861.			1871.		
	Really older.	Age correct.	Really y'nger.	Really older.	Age correct	Really y'nger.	Really older.	Age correct.	Really y'nger.
0-5	...	1000	...	...	100.0	...	...	100.0	...
5-10	...	97.0	3.0	...	97.1	2.9	...	97.3	2.7
10-15	...	98.8	1.2	...	98.3	1.7	...	98.3	1.7
15-20	...	99.1	.9	...	98.3	1.7	...	98.2	1.8
20-25	1.0	98.7	.3	.9	98.4	.7	.5	98.6	.9
25-30	1.6	98.4	...	1.0	99.0	...	.6	99.4	...
30-35	.7	99.3	...	.1	99.9	...	...	100.0	...
35-40	.2	99.8	...	...	100.0	...	...	100.0	...
40-45	3.5	96.5	...	1.6	97.8	.6	1.0	98.3	.7
45-50	5.0	95.0	...	1.9	98.1	...	1.2	98.8	...
50-55	9.0	91.0	...	4.2	95.8	...	3.1	96.9	...
55-60	4.7	95.3	...	1.2	98.8	...	.6	99.4	...
60-65	7.2	92.8	...	4.9	95.1	...	4.1	95.9	...
65-70	...	100.0	...	...	100.0	...	...	100.0	...
70-75	...	100.0	...	...	100.0	...	...	100.0	...
75-80	...	97.3	2.7	...	97.9	2.1	...	98.8	1.2
80-85	...	88.9	11.1	...	91.2	8.8	...	94.8	5.2
85-90	...	87.3	12.7	...	89.4	10.6	...	94.9	5.1
90-95	...	86.3	13.7	...	88.6	11.4	...	93.7	6.3
95-100	...	57.0	43.0	...	59.9	40.1	...	74.4	25.6
100 & up.	...	25.6	74.4	...	36.4	63.6	...	41.5	58.5

The emigration ^[15] at several ages shewn in Table VI, though graduated with much regularity, is very different in amount at particular ages from that which has been computed on page 14 (*ante*), and each person must form his own opinion as to which set of figures is likely to be nearest the truth.

Applying the final test previously used, by calculating the ratios of survivors indicated by the corrected figures, we have:—

Age at the end of the ten years.	Proportion of survivors (including emigrants) out of 100 males who were living ten years earlier.		The like proportion according to the English Life Table No. 3.
	According to corrected figures, 1851 and 1861.	According to corrected figures, 1861 and 1871.	
10-15	87.1	87.0	86.4
15-20	94.3	94.8	94.2
20-25	93.7	94.1	93.7

25–30	92.0	92.0	92.0
30–35	91.4	91.1	90.9
35–40	90.6	89.8	89.9
40–45	89.3	88.3	88.6
45–50	87.4	86.4	86.8
50–55	84.9	84.3	84.3
55–60	81.0	80.2	80.5
60–65	76.6	75.1	75.3
65–70	68.8	66.5	67.7
70–75	57.7	56.3	56.9
75–80	42.9	40.8	43.5

These ratios, like those obtained from the corrected female population, shew a great deal of regularity, and resemble those derived from the English Life Table very closely, whilst they deviate widely from those based upon the uncorrected census figures.

It may perhaps be supposed that such resemblance is artificial, and is really the result of the adoption of the Life Table as a guide in the apportionment of the recorded deaths under the years of birth. I am, however, sure that such a use of the Life Table cannot have controlled the result to any very important extent. Any apportionment of deaths occurring amongst a gradually increasing population like that of England, effected on a consistent and reasonable plan, would necessarily come within a very few thousands of the figures shewn in Tables IV and VI, at least for that period of life extending over fifty years or thereabouts, which lies between childhood and old age.

[16]

There is this further remark to be made, viz., that the series of ratios, though they resemble those derived from the Life Table, deviate from them at particular ages to a very appreciable extent thus:—

Age at end of decennium.	MALES—Loss by death.			FEMALES—Loss by death.		
	Life Table.	Corrected Population, 1851/60.	Corrected Population, 1861/70.	Life Table.	Corrected Population, 1851/60.	Corrected Population, 1861/70.
25–30	8.0	8.0	8.0	8.4	8.6	8.1
30–35	9.1	8.6	8.9	9.6	9.5	9.0
35–40	10.1	9.4	10.2	10.4	10.0	9.9
40–45	11.4	10.7	11.7	11.4	11.0	10.9
45–50	13.2	12.6	13.6	12.6	11.7	11.8
50–55	15.7	15.1	15.7	14.1	13.2	13.2
55–60	19.5	19.0	19.8	16.8	15.3	15.4
60–65	24.7	23.4	24.9	21.8	19.8	20.5

The actual mortality seems in general to be lower than that shewn in the Life Table, sometimes to the extent of five or six or even nine per cent. If, therefore, we were to compute the numbers of deaths on the basis of the Life Table, the result would be found to exceed the recorded deaths by many thousands. I naturally prefer to accept the teachings of the recorded facts, although they may not have been transposed quite correctly, rather than rely upon the Life Table,—which I feel sure has been graduated by some mathematical process at least as empirical as any estimate of mine. At the same time, I think I am bound to point out that so near a correspondence between the general character of my results and that of those obtained by mathematical graduation is a most important

fact, tending to convince us more strongly than ever that great regularity would be found to exist in the age-distribution of deaths occurring amongst a large population, and during a moderately long period of time, if only a truthful record of ages could be secured.

APPLICATION OF THE SUGGESTED CORRECTIONS.

On applying to the census figures of 1861 and 1871 for each of the eleven divisions, the proportional corrections shewn in Statements A and B, certain results were obtained, of which the following is an example:—

Age in 1871.	Division VIII (North-Western). Female Population in		Ratio of Population in 1871 to that in 1861, (per cent.)	National ratio of survivors (including allowance for Emigrants.)	Difference.
	1861.	1871.			
10–15	205692	179947	87.5	87.5	...
15–20	167248	171382	102.5	94.7	+7.8
20–25	151238	155554	102.9	93.7	+9.2
25–30	149921	145825	97.3	91.9	+5.4
30–35	144649	131174	90.7	91.0	-.3

The final result of the above calculation is a column of differences which, if the rate of mortality in Lancashire and Cheshire exactly equalled that of the nation, would represent the gain or loss at each age on a balance of migrations. The whole of the differences thus ascertained for the eleven divisions respectively are shewn in Table VIII.

The last column in that table, shewing the differences for England and Wales, of course represents simply the effect of migrations. I think that the differences in the other columns, at ages up to 35, are almost wholly consequent upon migrations. ^[196] The mortality in London and in Lancashire being greater than the average, the figures at those ages are perhaps less striking (because partially neutralised by such excessive mortality) than if the results of migrations stood out by themselves. It seems clear, in fact, that whilst the South-western counties *lose* more than 26.8 per cent. of their young men in the ten years beginning with age 10 to 15 and ending with age 20 to 25, London *gains* rather more than 14.5 per cent. at the same time of life.

The ratios last mentioned, and many others which are shewn in Table VIII, are of great importance, as indicating the movements of large numbers of persons, ^[198] and therefore, by way of rendering our impressions about their meaning more definite, I have taken the pains to apportion the deaths recorded in Divisions I, V, and VIII at certain ages, with the following results:—

Born in.	1861.	1871.	Apportioned Deaths, 1861–70.	Loss or gain by migrations.	Per cent. on Population in 1861.		The per centages in Table VIII being consequently made up thus—		
					Deaths	Loss or Gain by Migrations.	Loss or gain compared with average death loss. ^[20]	Loss or gain by migrations.	Total.
Male Population (corrected.)									
Div. I. LONDON.									
1851–55	147228	141937	7849	+2558	5.3	+1.7	-.1	+1.7	+1.6
1846–50	130615	141809	8042	+19236	6.1	+14.7	-.2	+14.7	+14.5

1841-45	118767	134948	11004	+27185	9.3	+22.9	-1.3	+22.9	+21.6
1836-40	120587	118776	12405	+10594	10.3	+8.8	-1.4	+8.8	+7.4
Div. V. SO.-WEST.									
1851-55	106614	91014	4601	-10999	4.3	-10.3	+9	-10.3	-9.4
1846-50	100897	67943	4838	-28116	4.8	-27.9	+1.1	-27.9	-26.8
1841-45	96505	57468	5637	-27400	6.2	-30.3	+1.8	-30.3	-28.5
1836-40	69223	50745	5430	-13048	7.8	-18.9	+1.1	-18.9	-17.8
Div. VIII. LANC. AND CHESH.									
1851-55	166782	160706	10641	+4565	6.4	+2.7	-1.2	+2.7	+1.6
1846-50	150583	145788	10945	+6150	7.3	+4.1	-1.4	+4.1	+2.7
1841-45	138424	133781	13247	+8604	9.6	+6.2	-1.6	+6.2	+4.6
1836-40	132498	119061	13348	-89	10.1	-1	-1.2	-1	-1.2
Female Population (corrected.)									
Div. I. LONDON.									
1851-55	149084	164132	7810	+22858	5.2	+15.3	+1	+15.3	+15.4
1846-50	133936	165675	7908	+39647	5.9	+29.6	+4	+29.6	+30.0
1841-45	139844	155003	10469	+25628	7.5	+18.3	+6	+18.3	+18.9
1836-40	143074	136729	11944	+5599	8.3	+3.9	+7	+3.9	+4.6
Div. V. SO.-WEST.									
1851-55	106074	90500	4892	-10682	4.6	-10.1	+7	-10.1	-9.4
1846-50	97784	77303	5375	-15106	5.5	-15.4	+8	-15.4	-14.6
1841-45	91581	68751	6249	-16581	6.8	-18.1	+1.3	-18.1	-16.8
1836-40	77717	61231	5950	-10536	7.7	-13.5	+1.3	-13.5	-12.2
Div. VIII. LANC. AND CHESH.									
1851-55	167248	171382	10115	+14249	6.0	+8.5	-.7	+8.5	+7.8
1846-50	151238	155554	11094	+15410	7.3	+10.2	-1.0	+10.2	+9.2
1841-45	149921	145825	14024	+9928	9.3	+6.6	-1.2	+6.6	+5.4
1836-40	144649	131174	14900	+1425	10.3	+1.0	-1.3	+1.0	-.3

Similar tables might be constructed for every age, and not only for each registration division, but for every registration district in the kingdom.

It will be observed that the apparent gain of the metropolitan division through migrations is less striking than might have been expected, although it is very large. But what is really shewn is the *balance* resulting, after deducting from the *gain* of strangers, the *loss* arising from the removal of families over the border of the division into extra metropolitan Middlesex, Surrey and Kent, or even into the nearer parts of Essex and Hertfordshire. If the London boundary were largely extended, it would be found that the *gain* by immigration from a distance is larger, and the *loss* by emigration is less, than now appears; and, in short, the statements whether of urban gain or of rural loss at ages up to 35 would be more striking than those exhibited in Table VIII.

After 35, both sexes in London and in the north-western counties exhibit a steady loss at each age, no doubt attributable in the main to the heavy mortality experienced in those divisions. The rural divisions numbered V and XI shew a loss until past the age of 50, due to emigration. These and several other agricultural divisions

(those numbered II, III, and IV), shew considerable gains at the higher ages, partly due to their mortality being low, and partly resulting from other causes.

It is evident that those who emigrate beyond sea (from Division V for example) are older persons than those who leave their native division to seek employment at a short distance, as do the majority of those who migrate from the eastern counties (Division IV). Two-thirds of these latter are perhaps between the ages of 14 and 20 years when they depart, and very few of them can be more than 25 years old.

The apparent relative mortality of the sexes at certain ages must be influenced by the dissimilar degrees of inaccuracy in the population returns for males and females respectively, as the following short statement will shew:—

	Mean population 1861–71, from uncorrected figures.		Mean population 1861–71, from corrected figures.		Deaths 1861–70.		Deaths per 1000.			
							From uncorrected.		From corrected.	
	Males.	Fem'l's.	Males.	Fem'l's.	Males.	Fem'l's.	M.	F.	M.	F.
15–20	1021321	1035205	1011321	1035632	62921	68553	6.2	6.6	6.2	6.6
20–25	906063	1011063	892063	938433	76591	80463	8.5	8.0	8.6	8.6
25–30	788782	886088	788782	849341	147734	160329	9.9	9.7	9.9	10.0
30–35	704005	769381	710005	761546						

The facts I have stated, and the experiments I have made, are perhaps sufficient to suggest by what methods of estimation an idea can be gained of the distribution of population as to ages, when once the total increase or decrease is known. ^[22] But they also tend to shew the difficulties which surround the subject, and the need which exists that enquirers should summon up sufficient courage to treat with a certain degree of freedom the returns of the census and registration offices.

TABLE I.—Showing the Rate of Increase of Population (per cent.) of each sex and *at each age* in the several Registration Divisions of England and Wales during the ten years, 1851–61.

TABLE II.—Shewing the Proportion (per cent.) of Population enumerated in 1861 to that enumerated *at corresponding ages ten years earlier*, in the several Registration Divisions, and in England and Wales.

[These two Tables having been calculated simply with the object of shewing that the proportional results obtainable by a comparison of the numbers enumerated at successive censuses, at particular ages, do not display sufficient regularity to justify the belief that such proportions would be approximately maintained decennium after decennium,—it is considered unnecessary to print them. The remaining Tables are printed in full, as without reference to, and careful consideration of, the facts they display, the paper would almost lose its significance.]

TABLE III.—Shewing the enumerated population in 1851, 1861, and 1871, at each quinquennial period of life, the estimated births in 1851–70, the registered deaths apportioned according to date of birth, and the numbers of inhabitants lost or gained, on the hypothesis of the correctness of the preceding figures.

Born in	Population enumerated.			Deaths registered.		Difference—referable to errors and migrations ^[23a]			
	MALES.			MALES.		1851–60.		1861–70.	
	1851.	1861.	1871.	1851–60.	1861–70.	Loss.	Gain.	Loss.	Gain.
1866–70	...	2011024 ^[23b]	1536464	...	427200	...	...	47360	...
1861–65	...	1887702 ^[23b]	1350819	...	546170	...	...	...	9287

1856–60	1751531 ^[23b]	1354907	1220770	365536	180534	31088	...	...	46397
1851–55	1651656 ^[23b]	1172960	1084713	482227	60259	...	3531	27988	...
1846–50	1176753	1059889	951917	156291	62499	...	39427	45473	...
1841–45	1050228	957930	843278	58497	75494	33801	...	39158	...
1836–40	963995	860210	746320	60004	75606	43781	...	38284	...
1831–35	873236	734287	640819	69604	74657	69345	...	18811	...
1826–30	795455	661690	590097	67451	77910	66314	...	...	6317
1821–25	699345	590280	506947	65694	81085	43371	...	2248	...
1816–20	617889	551058	455788	66739	84309	92	...	10901	...
1811–15	532680	453310	345907	67483	89886	11887	...	17517	...
1806–10	474211	392196	294675	69394	95736	12621	...	1785	...
1801–05	392882	299000	205370	73888	103431	19994	...	...	9801
1796–1800	346104	265536	149887	78530	108473	2038	...	7176	...
1791–95	254892	175538	82091	84399	108450	...	5045	...	15003
1786–90	227240	128428	38573	90915	94795	7897	...	...	4940
1781–85	151640	71780	11685	92953	64919	...	13093	...	4824
1776–80	114730	34256	2383	83815	33036	...	3341	...	1163
1771–75	65016	10359	390	58972	11790	...	4315	...	1821
1766–70	31690	2191	41	30694	2768	...	1195	...	618
1761–65	10423	399	...	11270	397	...	1246	2	...
1756–60	2282	55	...	2781	25	...	554	30	...
1751–55	456	...	...	463	...	...	7	...	...
1746–50	78	...	...	28	...	50	...	...	...
Age not stated	...	...	...	908	...	...	908	...	...
Totals	12184412	13674985	11058934	2138536	2459489	342279	72662	256733	100171
	FEMALES.			FEMALES.					
1866–70	...	1936784 ^[23c]	1534812	...	359329	...	...	42643	...
1861–65	...	1814081 ^[23c]	1355707	...	473449	...	...	...	15075
1856–60	1681961 ^[23c]	1345875	1203469	305511	172128	30575	...	...	29722
1851–55	1586949 ^[23c]	1171106	1095699	419780	61093	...	3937	14314	...
1846–50	1171354	1045287	1052843	151062	66098	...	24995	...	73654
1841–45	1042131	974712	937299	60003	79459	7416	...	...	42016
1836–40	949362	969283	813675	65106	81018	...	85027	74590	...
1831–35	883953	834877	700534	76754	79463	...	27678	54880	...
1826–30	871152	725088	639705	76237	77870	69827	...	7513	...
1821–25	771130	634262	546094	73309	76696	63559	...	11472	...
1816–20	658237	583069	488901	70730	75357	4438	...	18811	...

1811–15	555879	477530	372261	67174	77049	11175	...	28220	...
1806–10	494408	414367	328010	63923	84997	16118	...	1360	...
1801–05	406107	315004	235868	65030	97481	26073	...	...	18345
1796–1800	362697	290704	174086	72028	108636	...	35	7982	...
1791–95	271395	201034	99896	82975	114233	...	12614	...	13095
1786–90	254070	152917	51265	93843	105704	7310	...	...	4052
1781–85	175879	88860	17896	99612	78080	...	12593	...	7116
1776–80	135432	45403	4338	94274	43589	...	4245	...	2524
1771–75	81086	15608	855	71487	17466	...	6009	...	2713
1766–70	42150	3994	119	40514	4849	...	2358	...	974
1761–65	14982	839	...	16604	894	...	2461	...	55
1756–60	3969	146	...	4724	73	...	901	73	...
1751–55	874	...	...	921	...	...	47	...	...
1746–50	137	...	...	76	...	61	...	...	...
Age not stated	...	...	...	502	...	...	502	...	...
Totals	12415294	14040830	11653332	2072179	2386011	236552	183402	261858	209371

TABLE IV.—Shewing the Female population in 1851, 1861, and 1871, as corrected upon certain hypotheses, the estimated births in 1851–70, the registered deaths, apportioned according to date of birth, after adjustment, and the loss or gain of inhabitants upon a balance of migrations.

Born in	Population (corrected).			Deaths registered.		Loss or gain by migrations.			
	FEMALES.			FEMALES.		1851–60.		1861–70.	
	1851.	1861.	1871.	1851–60. ^(24b)	1861–70.	Loss.	Gain.	Lose.	Gain.
1866–70	...	1936784 ^(24a)	1571448	...	359329	...	...	6007	...
1861–65	...	1822952 ^(24a)	1340794	...	473449	...	...	8709	...
1856–60	1690145 ^(24a)	1379277	1203469	305511	172128	5357	...	3680	...
1851–65	1586949 ^(24a)	1157052	1094603	419780	61093	10117	...	1350	...
1846–50	1203052	1046332	981249	151062	66098	5658	...	...	1015
1841–45	1028583	976661	898871	60003	79459	...	8081	...	1669
1836–40	954109	895618	809607	65106	81018	...	6015	4993	...
1831–35	887489	799812	711042	76754	79463	10923	...	9307	...
1826–30	804073	713486	624991	76237	77870	14350	...	10625	...
1821–25	736430	652021	567938	73309	76696	11100	...	7387	...
1816–20	645730	569658	488901	70730	75357	5342	...	5400	...
1811–15	573667	501406	421400	67174	77019	5087	...	2957	...
1806–10	483036	415610	328010	63923	84997	3503	...	2603	...
1801–05	426412	357844	258275	65030	98802	3538	...	767	...
1796–1800	364148	290704	177568	72028	112636	1416	...	500	...

1791-95	308305	222745	104192	84226	118269	1334	...	284	...
1786-90	254070	156434	49008	97626	107225	10	...	201	...
1781-85	195578	92680	17144	103438	70172	...	540	...	636
1776-80	138547	43269	4121	95728	39172	...	450	...	24
1771-75	84572	14952	641	69678	14330	...	58	...	19
1766-70	40043	3734	60	36401	3732	...	92	...	58
1761-65	14188	614	...	13613	617	...	39	...	3
1756-60	3671	56	...	3631	50	...	16	6	...
1751-55	629	...	...	636	...	...	7	...	...
1746-50	52	...	...	53	...	...	1	...	...
Totals	12423478	14049701	11653332	2071677	2335011	77735	15899	64782	3424

TABLE V.—Shewing the degree of incorrectness of the returns of the ages of the female population, according to the hypotheses upon which Table IV is based.

	1851.			1861.			1871.		
True Age.	Returned at next lower age.	Returned correctly.	Returned at next higher age.	Returned at next lower age.	Returned correctly.	Returned at next higher age.	Returned at next lower age.	Returned correctly.	Returned at next higher age.
0-5	...	1171354	31698	...	1345875	33402	...	1534812	36636
5-10	...	1010433	18150	...	1137704	19348	...	1319071	21723
10-15	...	931212	22897	...	1025939	20393	...	1181746	21723
15-20	...	861056	26433	...	954319	22342	...	1073976	20627
20-25	...	804073	...	...	895618	...	...	981249	...
25-30	40646	695784	...	51323	748489	...	50967	847904	...
30-35	75346	570384	...	86388	627098	...	89395	720212	...
35-40	87853	485814	...	97990	554031	...	93463	617579	...
40-45	70065	412971	...	80231	489427	...	82955	542036	...
45-50	81437	344975	...	93642	407764	...	97669	470269	...
50-55	61132	303016	...	69766	345844	...	75825	413076	...
55-60	59681	248624	...	68523	289321	...	75325	345575	...
60-65	22771	231299	...	25683	265021	...	26686	301324	...

65–70	22771	172807	...	25683	197062	...	26686	231589	...
70–75	3072	135432	43	3972	152462	...	4279	173289	...
75–80	...	81043	3529	455	88860	3365	797	99896	3499
80–85	...	38621	1422	...	42038	1231	...	47766	1242
85–90	...	13560	628	...	14377	575	...	16654	490
90–95	...	3341	330	...	3419	315	...	3848	273
95–100	...	544	85	...	524	90	...	582	59
100–	...	52	...	...	56	...	...	60	...

TABLE VI.—Shewing the Male population in 1851, 1861 and 1871, as corrected upon certain hypotheses; the estimated births in 1851–70; the registered deaths apportioned according to date of birth, after adjustment; and the loss or gain of inhabitants upon a balance of migrations.

Born in	Population (Corrected.) Males.			Deaths Registered. Males.		Loss or gain by migrations.			
	1851.	1861.	1871.	1851–60. ^(25a)	1861–70.	1851–60.		1861–70.	
	1851.	1861.	1871.	1851–60. ^(25a)	1861–70.	Loss.	Gain.	Loss.	Gain.
1866–70	...	2006083 ^(25a)	1572464	...	427200	...	...	6419	...
1861–65	...	1892329 ^(25a)	1335819	...	546170	...	...	10340	...
1866–60	1758383 ^(25a)	1388307	1218770	365536	180534	4540	...	...	10997
1851–55	1651656 ^(25a)	1157960	1674713	482227	60259	11469	...	22988	...
1846–60	1208453	1057889	937917	156291	62499	...	5727	57473	...
1841–45	1030228	947930	843278	58497	75494	23801	...	29158	...
1836–40	960000	846210	751320	60004	75606	53786	...	19284	...
1831–35	868231	734287	644819	69604	74657	64340	...	14811	...
1826–30	784455	668690	580097	67451	77910	48314	...	10683	...
1821–25	696345	594280	506947	65694	81085	36371	...	6248	...
1816–20	624889	539058	447788	66739	84369	19092	...	6901	...
1811–15	535680	453310	357907	67483	89886	14887	...	5517	...
1806–10	458711	384196	284675	69394	95736	5121	...	3785	...
1801–05	389882	312000	217370	73888	104635	3994	...	...	10005
1796–1800	334904	256136	150887	78530	111999	238	...	...	6750
1791–95	273892	188538	83091	85504	111681	...	150	...	6234
1786–90	222840	129928	37173	94206	95445	...	1294	...	2690
1781–85	168040	73280	11235	96016	62880	...	1256	...	835

1776–80	116466	32356	2333	84496	29493	...	386	530 ^(25c)	...
1771–75	66800	9509	314	57119	9550	172	...	...	355 ^(25c)
1766–70	29493	2101	17	27390	2118	2	...	...	34
1761–65	9412	274	...	9116	266	22	...	8	...
1756–60	2166	20	...	2114	17	32	...	3	...
1751–55	318	...	...	310	...	8	...	...	...
1746–50	20	...	...	19	...	1	...	...	...
Totals	12191264	13674671	11058934	2137628	2459489	286190	8813	194148	37900

TABLE VII.—Shewing the degree of incorrectness of the returns of the ages of the Male population, according to the hypotheses upon which Table VI is based.

True Age.	1851.			1861.			1871.		
	Returned at next lower age.	Returned correctly.	Returned at next higher age.	Returned at next lower age.	Returned correctly.	Returned at next higher age.	Returned at next lower age.	Returned correctly.	Returned at next higher age.
0–5	...	1176753	31700	...	1354907	33400	...	1536464	36000
5–10	...	1018528	11700	...	1139560	18400	...	1314819	21000
10–15	...	952295	7705	...	1041489	16400	...	1199770	19000
15–20	...	865531	2700	...	941530	6400	...	1065713	9000
20–25	...	784455	...	...	846210	...	...	937917	...
25–30	8300	688045	...	7600	726687	...	5000	838278	...
30–35	11300	613589	...	7600	661090	...	5000	746320	...
35–40	4300	531380	...	600	590280	3400	...	640819	4000
40–45	1300	457411	...	...	539058	...	...	580097	...
45–50	16800	373082	...	8600	444710	...	6000	500947	...
50–55	19800	315104	...	8600	375596	...	6000	441788	...
55–60	31000	242892	...	16600	295400	...	14000	343907	...
60–65	12000	210840	...	3600	252536	...	2000	282675	...
65–70	16400	151640	...	13000	175538	...	12000	205370	...
70–75	...	114730	1736	...	128428	1500	...	149887	1000
75–80	...	63280	3520	...	70280	3000	...	81091	2000
80–85	...	28170	1323	...	31256	1100	...	36573	600
85–90	...	9100	312	...	9259	250	...	11085	150
90–95	...	1970	196	...	1941	160	...	2233	100
95–100	...	260	58	...	239	35	...	290	24
100–	...	20	...	...	20	...	...	17	...

TABLE VIII.—Shewing the differences between (1) The National percentage of persons surviving at each age in 1871, (including estimated loss by emigration in 1861–71) calculated on the numbers ten years younger enumerated in 1861; and (2) the ratios of inhabitants enumerated in each Division in 1871, compared with the respective populations ten years younger enumerated in 1861. The populations employed have first been corrected according to Statements A and B.

Age in 1871	I. London.	II. South Eastern.	III. South Midland.	IV. Eastern.	V. South Western.	VI. West Midland.	VII. North Midland.	VIII. North Western.	IX. Yorkshire.	X. Northern.	XI. Welsh.	England and Wales.
MALES. Excess or deficiency of the ratio of survivors, compared with National ratio.												
10– 15	-4.3	+7.6	+6.1	+1.9	-1.3	-.3	-.6	-.7	+2.8	+4.2	-.7	+0.8
15– 20	+1.6	+1.2	-8.8	-11.2	-9.4	-3.6	-6.0	+1.6	+3.6	+4.8	-3.6	-2.0
20– 25	+14.5	-4.0	-20.9	-25.3	-26.8	-10.0	+14.2	+2.7	+4.4	+12.8	-9.9	-6.4
25– 30	+21.6	-3.2	-14.1	-21.2	-28.5	-8.4	+13.0	+4.6	+6.0	+14.6	-13.1	-3.0
30– 35	+7.4	-2.8	-4.2	-6.6	-17.8	-5.9	-6.0	-1.2	+4.5	+5.5	-9.5	-2.3
35– 40	-1.6	+1	+1.0	-.3	-8.8	+4.4	-2.3	-3.4	+3.8	+2.0	-7.0	-2.0
40– 45	-3.6	+2	+1.8	+5	-4.9	-3.4	+1.0	-2.3	+3.7	+3.3	-7.2	-1.5
45– 50	-5.9	+1.7	+2.4	+1.9	+1.1	+1.8	+6	-3.5	+1.8	+2.9	-3.0	-1.1
50– 55	-7.2	+2.1	+2.1	+2.4	-.8	-2.1	+1.8	-4.9	+2.5	+2.0	-.6	-1.2
55– 60	-10.6	+5.5	+2.6	+2.7	+1.3	-1.8	+2.9	-6.7	+1.4	+9	+9	-1.3
60– 65	-8.5	+4.2	+4.8	+5.7	+1.3	-1.2	+2.5	-7.3	-1.7	+7	+6	-1.0
65– 70	-4.2	+10.5	+8.3	+10.5	+7.2	+3.6	+6.0	-3.9	+9	+1.8	+2.3	+3.2
70– 75	-5.4	+9.0	+7.8	+10.0	+7.7	+2.5	+6.5	-6.3	-.3	+2.2	+2.1	+2.6
75– 80	-4.8	+8.0	+6.5	+9.4	+7.1	+4.3	+5.5	-4.5	-1.3	+1.7	+6.6	+3.3
80– 85	-3.1	+4.4	+2.6	+6.1	+4.9	+2.8	+3.0	-3.4	-2.1	+5	+8.0	+2.1
85– 90	-.4	+2.8	+1	+3.5	+1.6	+1.4	+6	-2.2	-1.5	+2.3	+4.7	+1.1
FEMALES. Excess or deficiency of the ratio of survivors, compared with the National ratio.												
10– 15	-3.0	+5.5	+1.6	-.5	-2.9	-1.4	-1.9	...	+1.2	+26	-2.3	-.3

15–20	+15.4	-1.0	-7.1	-15.0	-9.4	+3.1	-6.9	+7.8	+3.9	+1.2	-7.8	-.1
20–25	+30.0	+1.5	-11.7	-20.3	-14.6	-4.9	-11.1	+9.2	+3.6	+8	-12.9	+1.1
25–30	+18.9	+8.0	-6.0	-12.5	-16.8	-5.3	-10.3	+5.4	+3.5	+4.6	-11.9	+1.1
30–35	+4.6	+7.9	-1.0	+5.5	-12.2	-4.8	-5.9	-.3	+3.7	+5.6	+5.8	-.7
35–40	+3.1	+5.3	+1.3	-2.5	-7.2	+2.2	-3.3	-2.9	+3.7	+4.4	-4.4	-1.3
40–45	-5.0	+4.2	+1.3	-.3	-4.3	-1.8	-1.8	-2.1	+2.5	+1.5	-5.8	-1.4
45–50	-6.4	+3.7	+2.1	+3	-1.1	-.9	-1.1	-2.4	+1.8	+1.4	-2.8	-1.1
50–55	-5.8	+4.1	+1.7	+3	+7	-1.3	+3	-4.1	+1.5	+2.3	-.2	-.9
55–60	-6.9	+5.7	+3.6	+1.2	+1	-.8	+1.2	-5.0	+2.1	+2.2	-.3	-.6
60–65	-4.7	+4.3	+3.7	+2.2	+1.1	-.6	+5	-4.9	-1.1	-.1	+2.1	-.5
65–70	-3.9	+3.7	+4.0	+5.5	+3.2	+5	-.1	-6.9	-2.3	-.5	+1.2	-.3
70–75	-4.0	+6.0	+4.1	+6.1	+3.5	+3	+1.0	-8.4	+3.0	-.8	+8	-.2
75–80	-3.2	+3.5	+7	+5.7	+3.0	+3	+6	-7.1	-4.6	-.4	+4.3	-.1
80–85	-2.4	+2.1	-.1	+4.3	+4	+5	-1.2	-5.7	+3.8	-2.0	+8.1	-.2
85–90	+1	+1.6	+2	+3.8	+1.5	+8	-.7	-2.6	-1.4	-.1	+4.7	+7

Memo.—If the mortality in each Division was exactly the same, these ratios would truly represent loss by emigration or gain by immigration; however, there is a good reason to think that at the higher ages the losses are (in Divisions I and VIII especially) caused by excessive mortality, and the gains in other Divisions are largely occasioned by the mortality therein being below the average.

FOOTNOTES.

[4] This should be read “0 and under 5.”

[6] These estimates have been made upon the basis of an apportionment derived from a consideration of the “English Life Table No. 3;” but I refrain from going into a detailed account of the process, for fear of rendering this paper more prolix and uninteresting than its character necessitates.

[8a] Net gain of incoming Irish, foreigners, &c., in excess of those departing. Net loss of English-born Emigrants, in excess of English returning home.

[8b] This is framed on the assumption that the migrations in each year were exactly alike in number.

[11a] It will be noticed that some of the ratios of inaccuracy attributed to the census figures of 1851 are lower than those for later censuses. These exceptions to the rule of decreasing inaccuracy might be removed without any very violent disturbance of the estimates shewn in Table IV, but it is thought scarcely requisite to do so.

[11b] The percentage would be 20.0 (more or less), except at high ages, for each year of error.

[14] Net Gain.

[15] A further slight modification in the estimates of unregistered male births will be noticed. It affects the estimated emigration in 1861–70 to the extent of about 10,000 persons,—making it 156,248 instead of 165,800.

[16] The following specimen calculation, shewing the ingredients which constitute the computed deaths in 1861–1870 amongst females born in

1851–1855, in 1816–1820, and in 1811–1815 respectively, will illustrate what I mean:—

Year of Death.	Born 1851–55.				Born 1816–20.			Born 1811–15.		
	Age 5–10.	Age 10–15.	Age 15–20.	Totals.	Age 35–45.	Age 45–55.	Totals.	Age 45–55.	Age 55–65.	Totals.
1861	6730	518	...	7248	6234	645	6879	6514	...	6514
1862	5259	1603	...	6862	4999	1994	6993	6669	...	6669
1863	4630	2915	...	7575	3688	3392	7080	6813	...	6813
1864	2452	3989	...	6441	2350	5366	7716	7686	...	7686
1865	678	4626	...	5304	785	6993	7778	7873	...	7873
1866	...	4499	620	5119	...	8048	8048	7413	844	8257
1867	...	2949	1786	4735	...	7547	7547	5443	2410	7853
1868	...	2274	3045	5319	...	7359	7359	3813	3983	7796
1860	...	1437	4351	5788	...	7859	7859	2466	5985	8451
1870	...	510	6192	6702	...	8098	8098	873	8264	9137
	19749	25350	15994	61093	18056	57301	75357	55563	21486	77049

[19a] The first line of Table VIII shews the effect of the removal of families from the first division to the semi-suburban districts in divisions II and III. Such families take with them a good many children; hence the loss to the metropolitan division, and the gain to the two divisions mentioned of both boys and girls.

[19b] The following statement may make this fact clearer:—

Division.	Births recorded 1841–45.	Population in 1871 born 1841–45.	Proportion as 100 to—
I. London	316037	289951	91.7
V. South-western	266860	126219	47.3

VIII. North-western	392151	279606	71.3
---------------------	--------	--------	------

The survivors, according to the English Life Table No. 3, should be about 62.2 per cent. The low rate of mortality in the South-western counties renders it certain that, but for migrations, the ratio of enumerated population to the corresponding births would be higher than 62.2 in that division; in the other divisions it would be lower.

[20] I am of opinion that the gain in this column, so far as regards London death-rates, is due to the departure of many women when in bad health, some of whom die in the country. But for the effect of such departures, the recorded deaths and the apparent net gain by migrations would reach higher numbers.

[22] The procedure for instance in 1881 might be:—Take the English population at each age in 1871 as already corrected, and introduce the numbers of births returned in 1876–80 and 1871–75 as the first two terms of the series, adding a reasonable allowance for non-registration. Deduct the deaths in 1871–80, duly apportioned under periods of birth. The gross number of either sex enumerated in 1881 being known, the net loss or gain by migrations can then be ascertained. Apportion this in somewhat similar proportions to those observed in 1851–60. The final results will exhibit approximately the distribution of population by ages in 1881.

[23a] It will be noticed that I have paid no regard to the interval of time between the census day and the January preceding, treating the exact decennium as being a near enough equivalent of the interval between census and census.

[23b] These numbers represent the births in the periods mentioned, *plus* an allowance for omissions, viz. $1\frac{3}{4}$ per cent. on births in 1866–70, and 2, $2\frac{1}{4}$, and $3\frac{1}{2}$ per cent. on births in the earlier periods respectively.

[23c] The correction adopted in the case of female births is rather larger than in that of male births. These figures are those returned, *plus* 2 per cent. on the births in 1866–70, and $2\frac{1}{4}$, $2\frac{3}{4}$, and 4 per cent. respectively on the births in the earlier periods.

[24a] Births, plus allowance for those unregistered, say 2 per cent. on births in 1866–70, and $2\frac{3}{4}$, $3\frac{1}{4}$, and 4 per cent. respectively on births in the earlier periods.

[24b] The few deaths “age not stated” are disregarded.

[25a] Births, *plus* allowance for those unregistered, say $1\frac{1}{2}$ per cent. on births in 1866–70, and $2\frac{1}{4}$, 2.65 and 3.5 per cent. respectively on births in the earlier periods.

[25b] The few deaths “are not stated” are disregarded.

[25c] These figures are, I think, improbable; I suppose the apportionment of deaths may be chiefly in fault.

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